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RESEARCH ASSIGNMENT 11

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1. Business Analytics refers to the statistical methods and computing technologies for processing, mining and visualising data to uncover patterns, relationships and insights that enable better business decision-making. This further involves using data with the purpose of solving business problems, monitoring the business' fundamentals, identifying new growth opportunities.

2. Data Analytics focuses on processing and interpreting raw data to uncover trends, patterns and insights using technical tools such as SQL and Python. Unlike Business Analytics, Data Analytics does not go the step further of solving business problems as it is more technical-centric. Business Analytics is more action-centric.

3. Business Intelligence (BI) refers to the procedural and technical infrastructure that collects, stores and analyses data produced by a company's activities. Business Analysts and Data Analysts use it to transform raw data into actionable insights using tools like dashboards and reports to help organisations make data-driven decisions.

4. Data-driven decision making - this is the practice of basing business decisions, strategies, and operational actions on analysing and interpreting actual data, facts and metrics. Basically using empirical evidence to make decisions.

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5. Big Data - refers to massive, complex and rapidly growing data sets in all forms (structured / unstructured), it is generally charactised by high volume and variety that requires specialised tools to analyse and unlock insights.

6. As part of the 4 Types of Data Analytics we have

- Descriptive Analytics: this type looks at data and desribes what has happened in the past such as past sales, historical performance.

7. - Diagnostic Analytics: takes a step further from being just desriptive and provides a deeper analysis and interogates things like "Why past sales performed the way did?" - also known as root cause analysis as it is concerned with the cause of historical performance.

8. - Predictive Analytics: this type takes data from historical events and into a machine learning model to not only identify key trends and patterns but it focuses on finding or predicting what will happen next based on current data.

9 - Prescriptive Analytics: beyond predictive analytics which provides a prediction of future pertomance, prescriptive analytics suggests various courses of action that can be taken and outlines what the potential implications for each would be which could help businesses take better decisions to avoid poor pertormance proactively.

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10. Data Literacy refers to the ability to read, work with, being able to analyse and communicate and interpret data.

SECTION 2

11. Structured data is highly organised data, typically easy to be stored in rows and tables and fits easily into databases.

12. Unstructured data is data that does not follow a predefined structure or specific format. Examples include images, videos, audio files etc., that are harder to store ~~in~~ in a database and harder to analyse as easily as structured data

13. A Data Warehouse is a centralised system used to store large volumes of structured data (databases) from multiple sources.

14. Data Mining is the process of analysing large datasets to discover patterns, trends and any relationships between the data.

15. Data Visualisation is the process of representing data visually through charts, maps, graphs, dashboards, in a way that makes the data easy to understand and patterns can be easily identified.

16. Dashboard is a live visual interface that displays key data and metrics in one place, in real time, showing multiple KPIs.

17. A metric is a measurable value used to track performance. eg. number of website clicks, revenue value etc.

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18. KPI is a specific type of metric that is directly tied to important business goals. Examples can include specifically defined target metrics such as monthly revenue targets or customer retention rate.

19. Benchmark is a reference point that is identified and used as a standard to compare performance by analysing how below or above the performance is to the specific reference point.

20. Segmentation refers to the process of dividing data into meaningful groups based on shared characteristics. Examples include grouping customers into specific age groups or location etc.

SECTION 3

21. Conversion Rate (CVR) refers to the percentage of users who complete a desired action such as clicking on an ad or making a purchase. The rate is calculated as the number of users who perform the pre-defined goal divided by the total number of visitors.

22. Customer Aquisition Cost (CAC) is the total expense spent by a business to acquire its customers. It is a metric used to evaluate profitability and return on investment.

23. Customer Lifetime Value represents how much a customer is expected to spend with a company, used to measure the total worth or profit that comes from the entire customer relationship from first to their last purchase from the business.

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24. Churn Rate is the rate at which customers stop doing business with an entity and used to measure the percentage of service subscribers who discontinue their subscriptions over a period of time.

25. Customer Retention Rate is the opposite of churn as this metric measures the rate of customers that remain with a company over a specific time period.

26. Net Promoter Score is a metric that defines customer experience and is calculated by surveying customers, measuring customer loyalty and their likelihood to promote a business

27. Average Order Value refers to the average amount spent per transaction that is calculated by dividing total revenue by the number of orders. This helps businesses understand the average amount customers spend per order over a specific time period.

28. Sales Funnel is a visual roadmap of the customer journey from the initial step of awareness to the last step of purchasing. It summarises the different touchpoints and interactions a prospective customer would experience when interacting with a business. For businesses it is important for them to nurture leads and enhance their interactions with potential customer to increase their conversion rates and reach sales purchases.

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29. SECTION 4

29. A Marketing Campaign is a long-term approach to promote a product or service of a business. The main purpose is typically to increase sales of a specific product or the goal could also be to improve or strengthen the brand image of a company. The marketing is also generally done through multiple media avenues.

30. Post-Campaign Analysis is the process of evaluating the performance of a marketing campaign and assessing what worked and what did not through analysis of results from every channel used in the campaign and companies and teams can assess what was most and least effective.

31. Incremental Revenue is the profit a business gains from a increase in revenue that stems from a campaign or personilisation efforts undertaken by a business.

32. Return on Investment is a financial metric used to measure the profitability of a business as it calculates the net profit against the total amount spent as investment into the business

33. Return on Ad Spend (ROAS) is another financial metric that specifically measures how much revenue a business gains for every dollar spent on ad campaigns. This metric is important for helping marketing teams to understand the effectiveness of their advertising in driving revenue.

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34. Click-Through Rate (CTR) determines how effectively an online advertisement enguges viewers by measuring the % of people who click on a specific ad (link, email, search result) after viewing it. This helps businesses understand how well keywords, ads and/or content, resonates with their target audience.

35. Bounce Rate, unlike CTR, speaks to the % of people who after clicking onto your homepage, 'bounce' or leave without engaging with the rest of your site but instead leave after viewing just one page.

36. A/B Testing is a method of comparing two versions of a web page or of a app against each other to determine which one performs better. Other terms are split testing or bucket testing which all refer to the same concept. The testing is done by measuring user engagement through a dashboard and analyse results and see which version is more effective among users.

37. Lead Generation is the process of attracting and nurturing potential customers who show interest in your products or service and guiding them into your sales pipeline with the intention of closing a sale.

38. Marketing Attribution is a way in which advertisers determine how their marketing tactics and methods contributed to sales and conversions. This is for the purpose of identifying the channels that inspired potential buyers to take action the most.

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